

can be tested in three major steps as explained below.

Software testing

This project uses Eigenfaces algorithm in OpenCV to perform face recognition. To use this algorithm, create a set of training data with pictures of faces that are allowed to trigger the relay.

Follow the steps given below:

Execute the following command to run capture-positives script to find a single face image:

```
$ sudo python capture-positives.py
```

Wait for some time and observe the terminal until you see Press Button instruction. Press S1 to capture your face image. If the script detects a single face, it will crop and save the training image in positive sub-directory.

If the script cannot detect a face, or detects multiple faces, error message 'Could not detect single face!'

Check the image in capture.pgm' will be displayed. It is recommended to maintain a distance of about 0.5 metres from the camera while taking a picture.

Press Ctrl + C to stop the script. Open capture.pgm file to view the last captured image.

Check the face in the database and train the face recogniser by running train.py code:

```
$ sudo python train.py
```

Training the face-recognition model on Raspberry Pi will take about ten minutes. Once training is complete, you will see mean.png and positive_eigenface.png files to visualise Eigenfaces of the model.

Now, test the face recogniser to recognise the face trained earlier. Execute the following command:

```
$ sudo python box.py
```

Observe the terminal to see Press Button instruction. Aim the camera at your face and press S1.

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

You should see a message 'Button pressed, looking for face..' on the terminal. After a few seconds, if the face is recognised, you will see the message 'Recognised face!' and relay RL1 will be energised.

You can connect an electrical load across the relay contacts at CON1. A 12V solenoid lock may be used as electrical load during testing. If your face is recognised or detected, relay is energised, solenoid lock is activated and the secured door gets opened. **EPY**



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